

Lecture 23.03: Types of Machine Learning: Unsupervised & Reinforcement Learning

 Python 3.10+

 Scikit-Learn 1.4+

Status Completed

 Language: English | [हिंदी / Hinglish](#) |  [Back to Lecture 23](#)

1. Unsupervised Learning

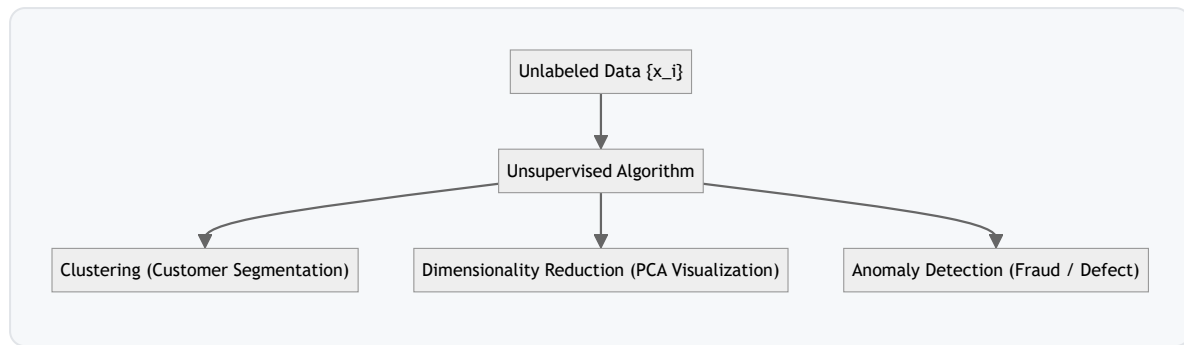
In **Unsupervised Learning**, the algorithm receives an **unlabeled dataset**:

$$\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^N$$

There are **no target labels** y_i and no supervisory feedback. The goal is to discover latent geometric structure, underlying distributions, or compressed representations.

Major Sub-Tasks in Unsupervised Learning:

1. **Clustering**: Partitioning data into K cohesive clusters based on distance metrics.
 - *Algorithms*: K-Means, DBSCAN, Hierarchical Clustering.
2. **Dimensionality Reduction**: Compressing d -dimensional feature spaces to $k \ll d$ latent dimensions while preserving variance.
 - *Algorithms*: Principal Component Analysis (PCA, Lecture 21), t-SNE, UMAP.
3. **Anomaly & Outlier Detection**: Identifying rare observations that deviate significantly from the baseline distribution.
4. **Density Estimation**: Modeling the probability density $p(\mathbf{x})$ (e.g., Gaussian Mixture Models).

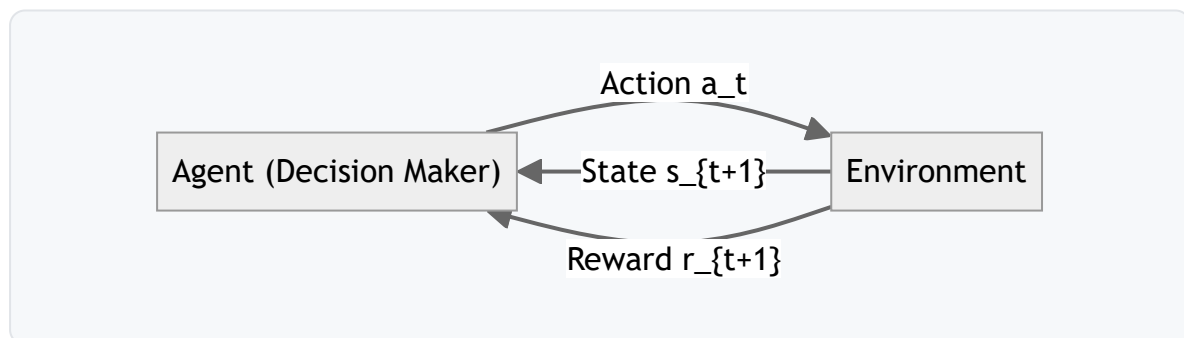


2. Reinforcement Learning (RL)

In **Reinforcement Learning**, an autonomous **Agent** learns to make sequential decisions by interacting with a dynamic **Environment** using trial-and-error to maximize cumulative scalar **Rewards**.

Reinforcement Learning is formulated mathematically as a **Markov Decision Process (MDP)** tuple (S, A, P, R, γ) :

- **State** ($s_t \in S$): Current situation of the agent.
- **Action** ($a_t \in A$): Decision chosen by the agent.
- **Reward** ($r_{t+1} \in \mathbb{R}$): Feedback signal evaluating the action.
- **Policy** ($\pi(a|s)$): The agent's decision-making strategy.





3. Master Comparison of Machine Learning Paradigms

Feature	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Data Nature	Labeled $(\mathbf{x}_i, y_i)$	Unlabeled $\mathbf{x}_i$	Environment States & Actions
Feedback Signal	Direct Ground Truth y_i	None (Self-organization)	Delayed Scalar Reward r_t
Objective	Predict output for new inputs	Discover latent patterns/clusters	Maximize cumulative reward $\sum \gamma^t r_t$
Key Algorithms	Linear/Logistic Reg, SVM, Trees	K-Means, PCA, GMM, Autoencoders	Q-Learning, PPO, Deep Q-Networks (DQN)
Typical Use-Cases	Price prediction, spam detection	Customer segmentation, compression	Robotics, Chess/Go, Autonomous driving

4. Python Demonstration: K-Means Clustering vs. Labeled Data

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Generate synthetic 2D points with 2 natural clusters
np.random.seed(42)
c1 = np.random.randn(50, 2) + np.array([2, 2])
c2 = np.random.randn(50, 2) + np.array([-2, -2])
X_unlabeled = np.vstack([c1, c2])

# Apply unsupervised K-Means clustering (no labels given!)
kmeans = KMeans(n_clusters=2, random_state=42, n_init=10)
cluster_labels = kmeans.fit_predict(X_unlabeled)

print("K-Means discovered centroids:\n", np.round(kmeans.cluster_centers_, 3))
```